

Fatal Pedestrian and Cyclist Crash Risk Factors in Oregon: A Spatial and Predictive Analysis (2015-2024)

1. Title and Team Metadata

Project Title

Fatal Pedestrian and Cyclist Crash Risk Factors in Oregon: A Spatial and Predictive Analysis
(2015-2024)

Team Members

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- Owner Product Lead, Data Engineering

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- Statistical Modeling, Machine Learning

Owner Product Lead

Simon Thompson

Peer Stakeholder Product Owners

Amaya Supancich-McCord

Aiyana Brown

Seira Ramchandani

Shared Theme

Data-driven prediction of community and urban outcomes.

Studio Session

Studio Session 2

Team Size

2-person team

2. Problem Framing and Research Questions

Pedestrians and cyclists remain among the most vulnerable users of the transportation network. While motor vehicle safety has improved substantially over recent decades, pedestrian and cyclist fatalities continue to represent a significant public health and transportation safety concern. Understanding where these fatalities occur and the environmental, roadway, and demographic factors associated with them can help transportation planners prioritize interventions and allocate limited safety resources more effectively.

This proposal is consistent with the project Vision and Mission established in the project charter. We seek to identify factors associated with fatal pedestrian and cyclist crashes and provide evidence-based recommendations for transportation safety improvements.

Primary Research Question

Which environmental, roadway, and temporal factors are most strongly associated with fatal pedestrian and cyclist crashes in Oregon between 2015 and 2024?

Secondary Research Question

Are fatal pedestrian and cyclist crashes disproportionately concentrated in areas with certain identifiable demographics (race, ethnicity, income, etc.) in Oregon between 2015 and 2024?

3. Impact and Stakeholders

Stakeholders

1. Pedestrians and Cyclists: Pedestrians and cyclists experience the highest levels of physical vulnerability during roadway collisions. Findings may help identify locations requiring targeted safety improvements.
2. Oregon Department of Transportation (ODOT): ODOT maintains statewide transportation safety initiatives and may use project findings to better understand risk factors associated with fatal pedestrian and cyclist crashes.
3. City/County Transportation Planners: Local planners may benefit from spatial analyses identifying high-risk features such as traffic device (STOP sign, YIELD sign, signal), roadway surface, sidewalk presence, etc. for both retroactive and proactive infrastructure investment.
4. Public Health and Community Organizations: Organizations focused on injury prevention, transportation equity, and community mobility may use findings to support advocacy and planning efforts.

Shared Theme Influence

The peer stakeholder group's shared theme is data-driven prediction of community and urban outcomes. Feedback from peer stakeholders will help ensure project priorities remain aligned with broader societal and policy implications rather than focusing exclusively on technical modeling performance.

4. Data Sources and Access Plan

Primary Dataset

Fatality Analysis Reporting System (FARS)

Source: <https://www.nhtsa.gov/research-data/fatality-analysis-reporting-system-fars>

Description: The Fatality Analysis Reporting System is maintained by the National Highway Traffic Safety Administration (NHTSA) and contains census-level records of all police-reported motor vehicle crashes in the United States involving at least one fatality.

Coverage: 2015-2024

Estimated Volume: Approximately five thousand pedestrian and cyclist fatality records within Oregon between 2015 and 2024.

Refresh Cadence: Annual

Approval Status: Pending submission

Supplementary Data Sources

American Community Survey (ACS)

Purpose: Neighborhood-level socioeconomic indicators including median household income, poverty rates, and population characteristics.

Refresh Cadence: Annual

Approval Status: Pending submission

5. Data Engineering Plan

The project will follow the DS3 repository structure.

data/	src/	notebooks/	deliverables/
• raw • interim • processed	• ingestion • cleaning • feature_engineering • modeling	• exploratory analysis • spatial analysis • modeling experiments	• poster • writeup • figures

Raw datasets will remain unchanged after acquisition. Processing scripts will generate reproducible interim and final datasets. All transformations will be documented through version-controlled Python scripts and notebooks.

Data dictionaries and metadata documentation will be maintained throughout the project lifecycle.

6. Planned Analysis

Create-Observe-Analyze PBI 1

Crash Data Preparation

Create: Build reproducible crash-data processing pipeline.

Observe: Generate cleaned analysis-ready dataset.

Analyze: Assess completeness, geographic coverage, and variable quality.

Create-Observe-Analyze PBI 2

Risk Factor Modeling

Create: Engineer environmental, roadway, temporal, and demographic features.

Observe: Fit statistical and machine learning models.

Analyze: Identify factors most strongly associated with fatal pedestrian and cyclist crashes.

Methods:

- Logistic Regression (where applicable)
- Random Forest
- Gradient Boosting / XGBoost (if feasible)

Evaluation:

- Cross-validation
 - Feature importance rankings
 - Model interpretability measures
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Create-Observe-Analyze PBI 3

Spatial Hotspot Identification

Create: Generate crash location maps.

Observe: Identify geographic clustering patterns.

Analyze: Determine whether fatal crashes concentrate in identifiable corridors or intersections.

Create-Observe-Analyze PBI 4

Transportation Equity Analysis

Create: Merge crash locations with census data.

Observe: Measure crash concentrations by neighborhood characteristics.

Analyze: Evaluate whether fatal crashes disproportionately occur in areas with certain identifiable demographics (race, ethnicity, income, etc).

Success Criteria

The primary question will be answered if the project identifies statistically meaningful roadway, environmental, or temporal factors associated with fatal pedestrian and cyclist crashes.

The secondary question will be answered if spatial and statistical analyses reveal measurable relationships between crash concentration and neighborhood characteristics.

7. Program Integration (Five Pillars)

Data Engineering

Construction of reproducible pipelines for acquisition, cleaning, integration, and documentation of multiple public datasets.

Analytics and Machine Learning

Use of statistical models and machine learning techniques to identify influential risk factors.

Visualization and Communication

Development of static maps, charts, and explanatory figures suitable for technical and non-technical audiences.

Ethics and Responsible Use

Evaluation of data limitations, fairness concerns, and potential consequences of policy recommendations.

Statistics and Research Design

Application of observational data analysis, hypothesis testing, regression methods, and validation strategies.

8. Ethics and Risk Notes

Underreporting and Coverage Bias

FARS includes only crashes involving at least one fatality. Findings therefore represent fatal crash risk and should not be generalized to all pedestrian or cyclist incidents.

Fairness and Equity

Historical crash reporting and transportation infrastructure investments may not be evenly distributed across communities. Findings must be interpreted carefully to avoid reinforcing existing inequities.

Privacy

The project uses publicly available administrative datasets containing no personally identifiable information.

Limitations

- Observational data cannot establish causality.
 - Limited sample size within the state of Oregon.
 - Potential missing contextual variables such as pedestrian volume or cyclist exposure.
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9. Timeline and Milestones

Weeks 4-6

Obtain and validate datasets

Build ingestion pipeline

Complete data cleaning

Conduct exploratory analysis

Week 7 – M2 Data Summary

Reproducible pipeline complete

Initial maps and descriptive statistics completed

Weeks 8-9

Feature identification / engineering

Preliminary modeling

Spatial hotspot analysis

Week 10 – M3 Poster Draft

Draft poster completed

Initial results and visualizations included

Weeks 11-12

Final modeling

Equity analysis

Interpretation of results

Week 12 – M4 Writeup Draft

Complete draft report submitted

Weeks 13-14

Refine visualizations

Finalize recommendations

Address peer feedback

Week 14 – M5 Final Submission

Final report

Final poster

Presentation-ready deliverables

Risks and Mitigations

Risk	Mitigation
Geocoding inconsistencies	Validate spatial joins and conduct quality checks
Schedule delays	Maintain milestone-tagged backlog and weekly reviews

10. DS3 Process and Tooling

GitHub Repository

<https://github.com/S1mT/data510-OregonPedSafety>

GitHub Projects Board

<https://github.com/S1mT/data510-OregonPedSafety/projects?query=is%3Aopen>

Discord Category

Project Discord category containing:

#general

#standup

#studio

#blockers

Charter Status

CHARTER.md has been committed and includes:

- Vision
- Mission
- Context
- Success Criteria

- Working Agreements
- Response SLAs
- Definitions of Ready and Done

Backlog Status

Milestone-tagged PBIs have been seeded and will continue to be refined throughout the semester.

Studio Engagement

We will participate in weekly Studio Brief and Studio Critique cycles, maintain Iteration Reviews within the repository README, and incorporate stakeholder feedback into backlog prioritization and project scope decisions.